

# Arithmetic and Order of Real Numbers

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- **Exercise 1**

Let  $a, b \in \mathbb{R}, b \neq 0$ . Prove that  $\frac{a}{b} > 0 \iff (a > 0 \wedge b > 0) \vee (a < 0 \wedge b < 0)$ .

Similarly, prove that  $\frac{a}{b} < 0 \iff (a > 0 \wedge b < 0) \vee (a < 0 \wedge b > 0)$ .

- **Proof 1.1**

Suppose  $a, b \in \mathbb{R}, b \neq 0$ .

**1. Proof of the first equivalence:**

- ( $\implies$ ) Suppose  $\frac{a}{b} > 0$ . By definition,  $\frac{a}{b} = a \cdot \frac{1}{b}$ . From the properties of ordered fields, a product of two numbers is positive if and only if both numbers have the same sign, which yields:

$$\left(a > 0 \wedge \frac{1}{b} > 0\right) \vee \left(a < 0 \wedge \frac{1}{b} < 0\right).$$

By Proposition 2.4, we know that  $\frac{1}{b} > 0 \iff b > 0$ , and similarly  $\frac{1}{b} < 0 \iff b < 0$ .

Substituting these relations into the statement, we have:

$$(a > 0 \wedge b > 0) \vee (a < 0 \wedge b < 0).$$

- ( $\impliedby$ ) Suppose  $(a > 0 \wedge b > 0) \vee (a < 0 \wedge b < 0)$ . We split this into two cases:

- **Case 1:** If  $a > 0$  and  $b > 0$ , then by Proposition 2.4,  $b > 0 \implies \frac{1}{b} > 0$ . Since the product of two positive numbers is positive, we obtain  $\frac{a}{b} = a \cdot \frac{1}{b} > 0$ .

- **Case 2:** If  $a < 0$  and  $b < 0$ , then by Proposition 2.4,  $b < 0 \implies \frac{1}{b} < 0$ . Since the product of two negative numbers is positive, we obtain  $\frac{a}{b} = a \cdot \frac{1}{b} > 0$ . Therefore, in both cases, we have  $\frac{a}{b} > 0$ .

**2. Proof of the second equivalence:**

- ( $\implies$ ) Suppose  $\frac{a}{b} < 0$ , where  $\frac{a}{b} = a \cdot \frac{1}{b}$ . From the properties of ordered fields, a product of two numbers is negative if and only if they have different signs, which yields:

$$\left(a > 0 \wedge \frac{1}{b} < 0\right) \vee \left(a < 0 \wedge \frac{1}{b} > 0\right).$$

Applying Proposition 2.4, we obtain:

$$(a > 0 \wedge b < 0) \vee (a < 0 \wedge b > 0).$$

- ( $\impliedby$ ) Suppose  $(a > 0 \wedge b < 0) \vee (a < 0 \wedge b > 0)$ . We consider the two cases:

- **Case 1:** If  $a > 0$  and  $b < 0$ , then by Proposition 2.4,  $\frac{1}{b} < 0$ . The product of a positive number and a negative number is negative, so  $\frac{a}{b} = a \cdot \frac{1}{b} < 0$ .

- **Case 2:** If  $a < 0$  and  $b > 0$ , then by Proposition 2.4,  $\frac{1}{b} > 0$ . The product of a negative number and a positive number is negative, so  $\frac{a}{b} = a \cdot \frac{1}{b} < 0$ . Therefore, the second equivalence is also proved.

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- **Exercise 2**

Let  $a, b \in \mathbb{R}$  satisfy  $0 < a < b$ . Prove that  $0 < \frac{1}{b} < \frac{1}{a}$ .

- **Proof 2.1**

Suppose  $a, b \in \mathbb{R}, 0 < a < b$ . Then  $0 < b - a$ . Therefore we have:

$$\frac{1}{a} - \frac{1}{b} = \frac{b - a}{ab}.$$

And since  $b - a > 0$  and  $ab > 0$  (as  $a > 0$  and  $b > 0$ ), we have that  $\frac{1}{a} - \frac{1}{b} > 0$ , that is,  $\frac{1}{a} > \frac{1}{b}$ . Because  $b > 0$ , by Proposition 2.4 we have  $\frac{1}{b} > 0$ . Therefore,  $0 < \frac{1}{b} < \frac{1}{a}$ .

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• **Exercise 3**

Let  $a, b$  be distinct positive real numbers. Prove that  $\frac{a}{b} + \frac{b}{a} > 2$ .

◦ **Proof 3.1**

Suppose  $a, b \in \mathbb{R}$  with  $a > 0, b > 0, a \neq b$ . By the properties of real numbers, we have that

$$(a - b)^2 > 0.$$

Then, since  $ab > 0$ , we expand and divide:

$$\begin{aligned} a^2 - 2ab + b^2 &> 0 \\ a^2 + b^2 &> 2ab \\ \frac{a^2}{ab} + \frac{b^2}{ab} &> \frac{2ab}{ab} \\ \frac{a}{b} + \frac{b}{a} &> 2. \end{aligned}$$


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• **Exercise 4**

Prove that  $\sqrt{6}$  is irrational.

◦ **Proof 4.1**

By contradiction. Suppose  $\sqrt{6}$  is rational. Then, there exist  $p \in \mathbb{Z}, q \in \mathbb{N}$  where  $p, q$  are coprime such that  $\sqrt{6} = \frac{p}{q}$ . Then

$$\begin{aligned} 6 &= \frac{p^2}{q^2} \\ 6q^2 &= p^2. \end{aligned}$$

Thus we have

$$\nu_3(6q^2) = \nu_3(6) + \nu_3(q^2) = \nu_3(p^2).$$

Using the properties of the prime exponent notation, this can be rewritten as  $1 + 2\nu_3(q) = 2\nu_3(p)$ . Since  $1 + 2\nu_3(q)$  is an odd integer and  $2\nu_3(p)$  is an even integer, this is a contradiction. Therefore,  $\sqrt{6}$  is irrational.

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• **Exercise 5**

Let  $x$  be irrational,  $r \in \mathbb{Q}$ . Prove that  $x + r$  and  $\frac{r}{x}$  when  $x \neq 0, r \neq 0$  are irrational.

◦ **Proof 5.1**

By contradiction. Suppose  $x \in \mathbb{R} \setminus \mathbb{Q}, r \in \mathbb{Q}$ . Suppose  $x + r \in \mathbb{Q}$ , then  $\exists s \in \mathbb{Q}$  such that  $x + r = s$ . Therefore,  $x = s - r$ . But  $s - r \in \mathbb{Q}$  because  $\mathbb{Q}$  is closed under subtraction. This contradicts the fact that  $x$  is irrational. Thus,  $x + r$  is irrational.

By contradiction. Suppose  $r \in \mathbb{Q}, x \neq 0, r \neq 0$ , and  $x$  is irrational. Suppose  $\frac{r}{x}$  is rational, then  $\exists s \in \mathbb{Q}$  such that  $\frac{r}{x} = s$ . Since  $r \neq 0$ , we have  $s \neq 0$ , which implies  $x = \frac{r}{s} = r \cdot s^{-1}$ . But since  $\mathbb{Q}$  is closed under division by nonzero elements,  $\frac{r}{s} \in \mathbb{Q}$ , which contradicts the fact that  $x$  is irrational. Therefore,  $\frac{r}{x}$  is irrational.

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• **Exercise 6**

Let  $n \in \mathbb{N}$  be odd. Prove that  $\forall x, y \in \mathbb{R}, x < y \implies x^n < y^n$ .

◦ **Proof 6.1**

Since  $n \in \mathbb{N}$  is odd, we can write  $n = 2k + 1$  for some  $k \in \mathbb{N}_0$ . We prove the statement by induction on  $k$ .

1. **Base step:** If  $k = 0$ , then  $n = 1$ . The implication  $x < y \implies x^1 < y^1$  is trivially true.
2. **Inductive step:** Suppose that the result is true for  $n = 2k + 1$ , that is,  $x < y \implies x^{2k+1} < y^{2k+1}$ . We want to show that it holds for  $2(k + 1) + 1 = n + 2$ . We can factor the difference as follows:

$$y^{n+2} - x^{n+2} = y^n y^2 - x^n x^2 = y^n(y^2 - x^2) + x^2(y^n - x^n) = y^n(y - x)(y + x) + x^2(y^n - x^n).$$

By the inductive hypothesis,  $x < y \implies y^n - x^n > 0$ , so the second term  $x^2(y^n - x^n) \geq 0$ . For the first term, we analyze three cases based on the signs of  $x$  and  $y$ :

- **Case 1:** If  $0 \leq x < y$ , then  $y^n > 0$  and  $y + x > 0$ . Since  $y - x > 0$ , the first term is strictly positive, hence  $y^{n+2} - x^{n+2} > 0$ .
- **Case 2:** If  $x < y \leq 0$ , then  $-y \geq 0$  and  $-x > -y \geq 0$ . By Case 1,  $(-x)^n < (-y)^n$ . Since  $n$  is odd,  $-x^n < -y^n \implies x^n > y^n$ . Also,  $y + x < 0$  and  $y - x > 0$ . Thus,  $y^n(y - x)(y + x) > 0$  (a product of two negatives and one positive), hence  $y^{n+2} - x^{n+2} > 0$ .
- **Case 3:** If  $x < 0 < y$ , then since  $n + 2$  is odd,  $x^{n+2} < 0$  and  $y^{n+2} > 0$ . It directly follows that  $x^{n+2} < 0 < y^{n+2}$ .

Therefore, in all cases,  $y^{n+2} - x^{n+2} > 0$ , which means  $x^{n+2} < y^{n+2}$ . By induction, the statement is true for all odd  $n \in \mathbb{N}$ .

• **Exercise 7**

Suppose  $x, y \geq 0, n \in \mathbb{N}$ . Prove that  $\sqrt[n]{xy} = \sqrt[n]{x} \cdot \sqrt[n]{y}$ .

◦ **Proof 7.1**

Suppose  $x, y \geq 0, n \in \mathbb{N}$ . Let  $u = \sqrt[n]{x}$  and  $v = \sqrt[n]{y}$ . Then  $u^n = x$  and  $v^n = y$ , where  $u, v \geq 0$ . Then we have  $(uv)^n = u^n v^n = xy$ . Since  $uv \geq 0$ ,  $uv$  must be the unique  $n$ -th root of  $xy$ . Thus,  $\sqrt[n]{xy} = \sqrt[n]{x} \cdot \sqrt[n]{y}$ .